Community Assembly II

Today's Agenda:

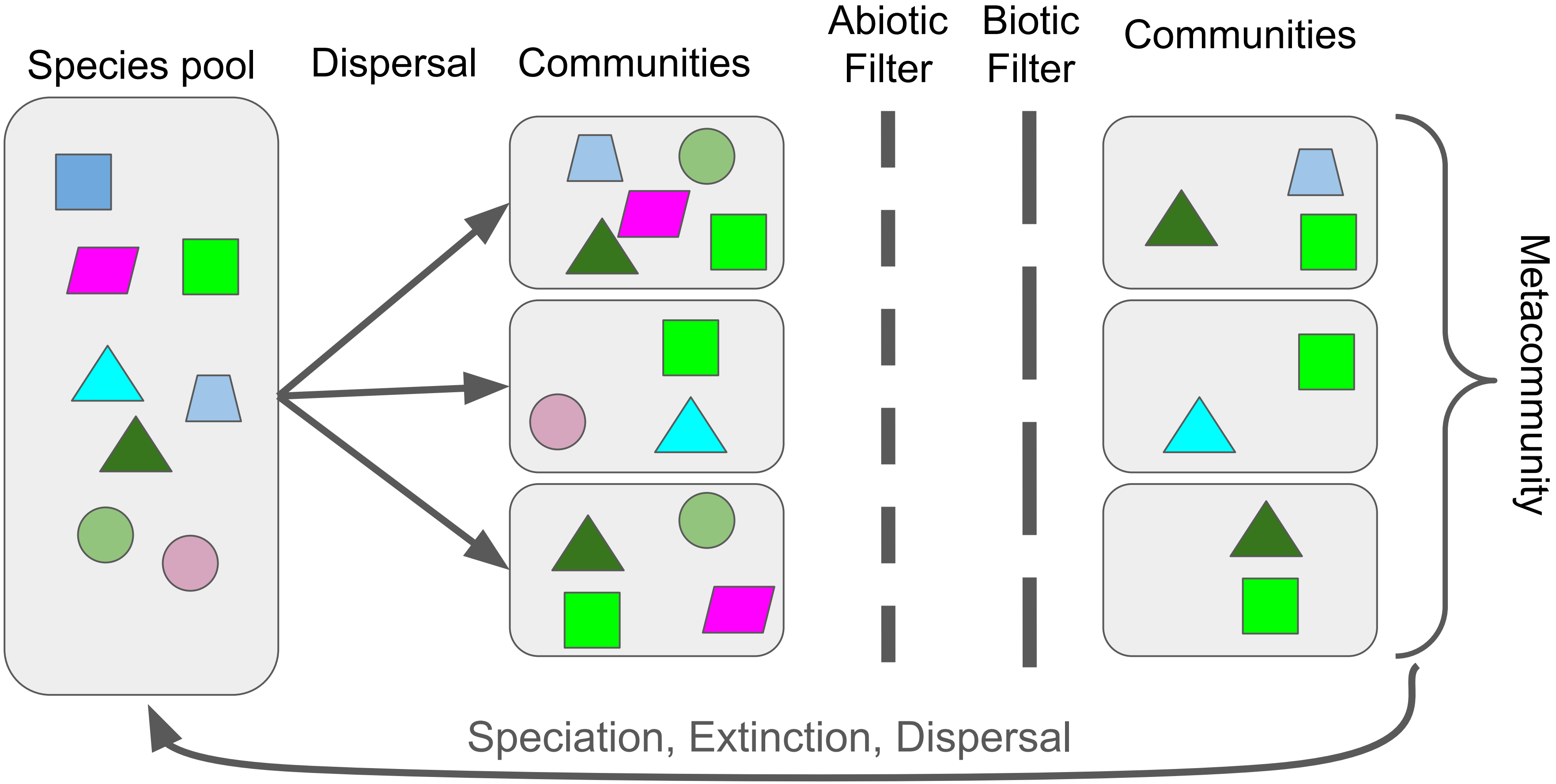
- Quiz
- Check-in
- Community Assembly Examples

Check-in

Just wanted to touch base and see how the class is going.

Anything that isn’t working? Anything that IS working?

Feel free to email me as well.


Species pool
Dispersal
Communities
Abiotic Filter
Biotic Filter
Communities
Metacommunity
Speciation, Extinction, Dispersal

What does this look like in practice?

Environmental Filtering of Plants

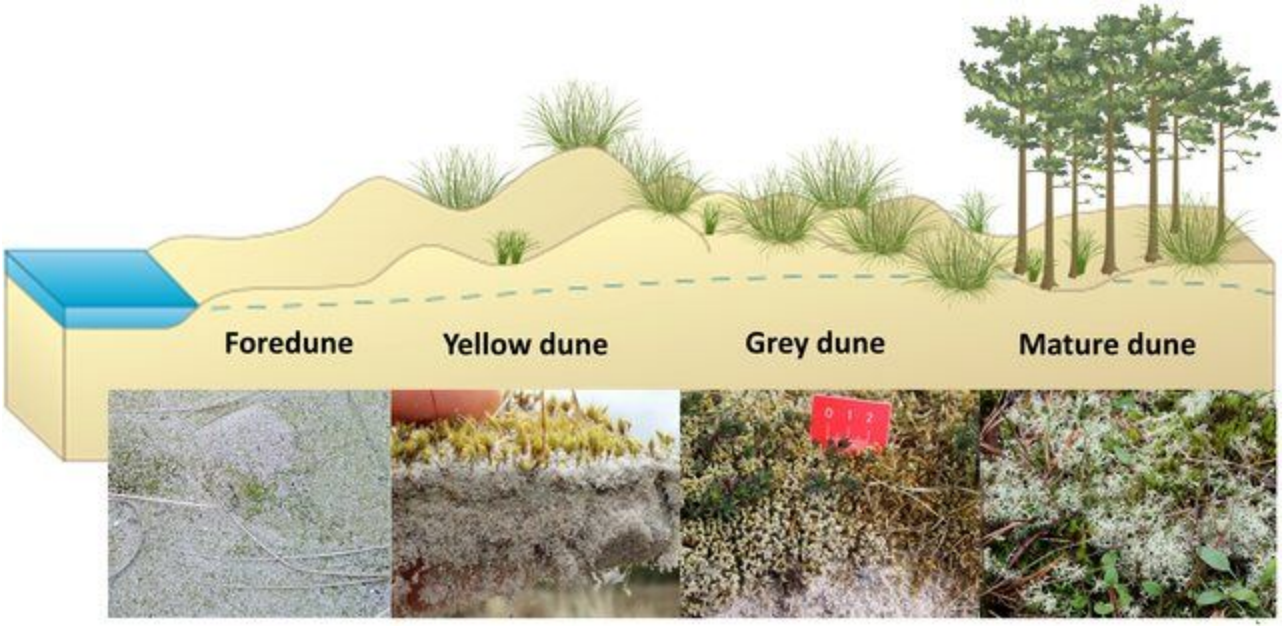
PLANT ECOLOGY

Environmental filtering explains variation in plant diversity along resource gradients

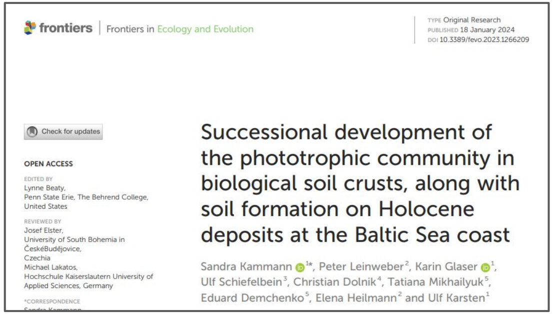
Etienne Laliberté,[1*] Graham Zemunik,[1] Benjamin L. Turner[2]

Studied a chronosequence of soil ages
- “Space-for-time” substitution
- Dunes in Australia

Aside: space for time substitutions



Dune Age



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Check for updates

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*CORRESPONDENCE

Successional development of the phototrophic community in biological soil crusts, along with soil formation on Holocene deposits at the Baltic Sea coast

Sandra Kammann[1*], Peter Leinweber[2], Karin Glaser[1], Ulf Schiefelbein[3], Christian Dolnik[4], Tatiana Mikhailyuk[5], Eduard Demchenko[5], Elena Heilmann[2] and Ulf Karsten[1]

- Note: this figure depicts Baltic dunes

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- “Space-for-time” substitution
- Dunes in Australia

Really cool system!

- 2 million year gradient of soil ages
- Sites within 10 km (assume the same regional species pool)
- Comparable climate
- Open ecosystem (not light-limited)

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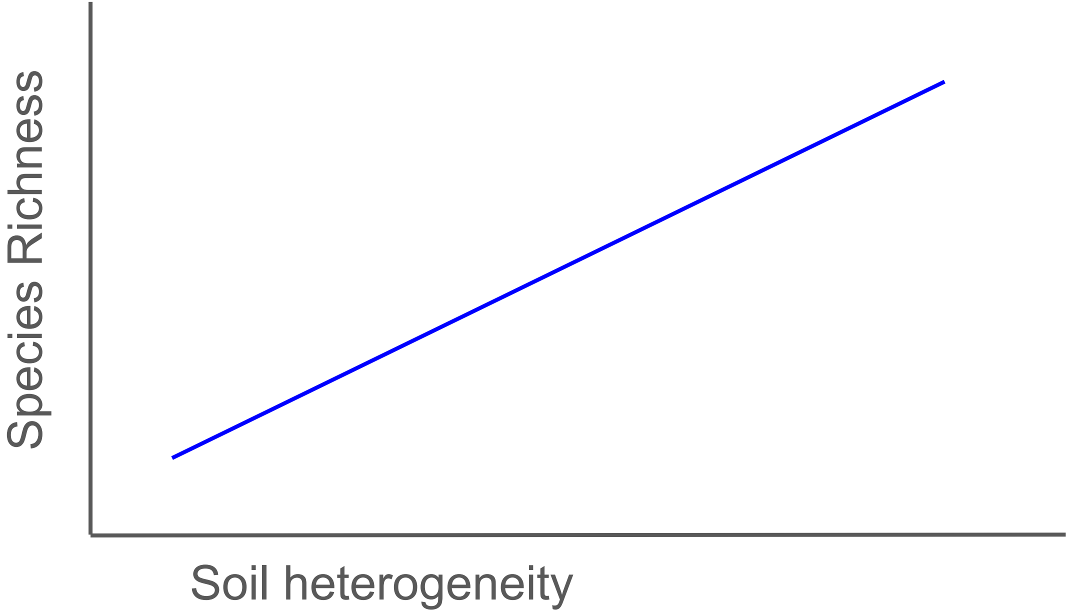
Hypotheses tested

More heterogeneous soils, more species - why?



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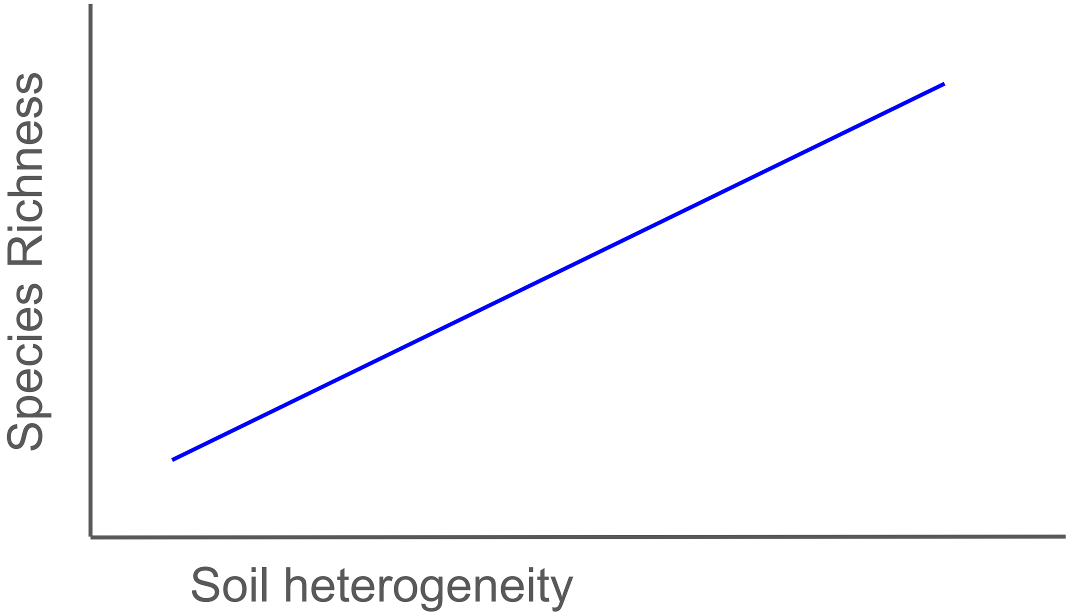
Hypotheses tested

More heterogeneous soils, more species (niche partitioning)



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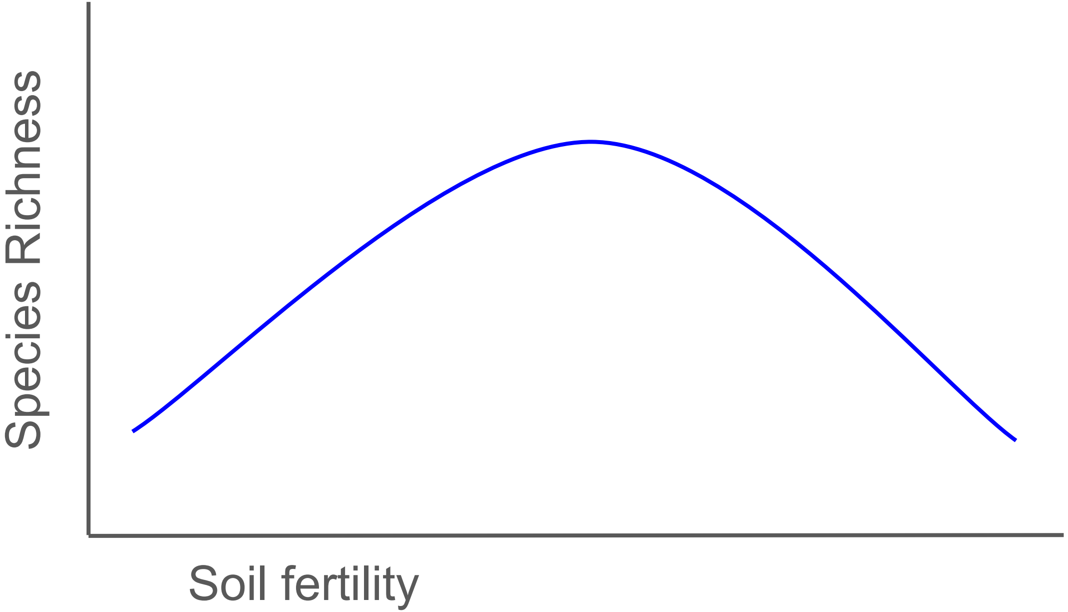
Environmental Filtering of Plants

Hypotheses tested

Unimodal response with soil fertility - why might we expect this?



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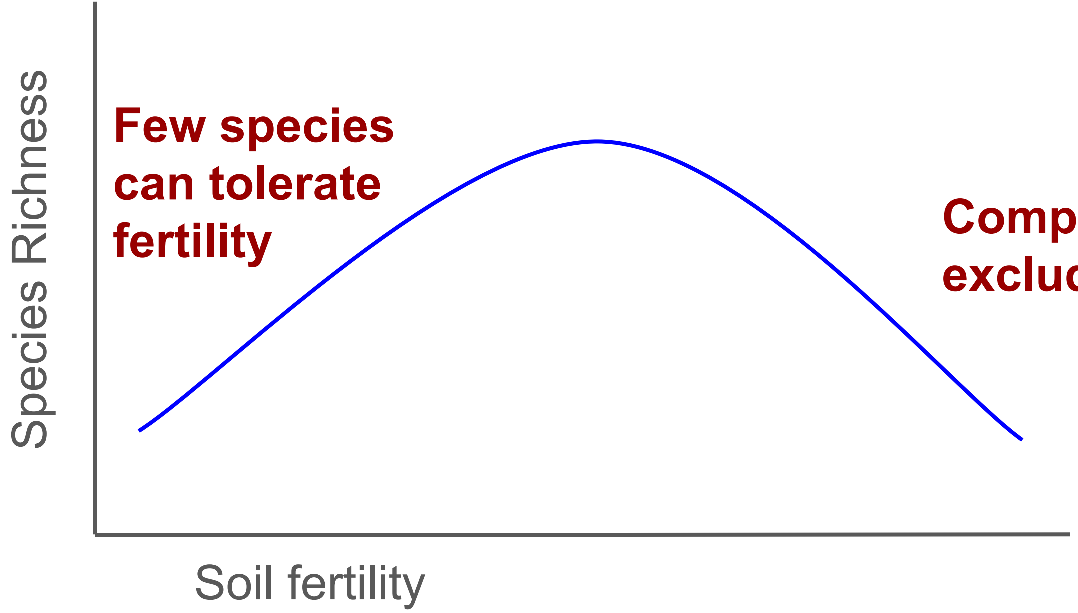

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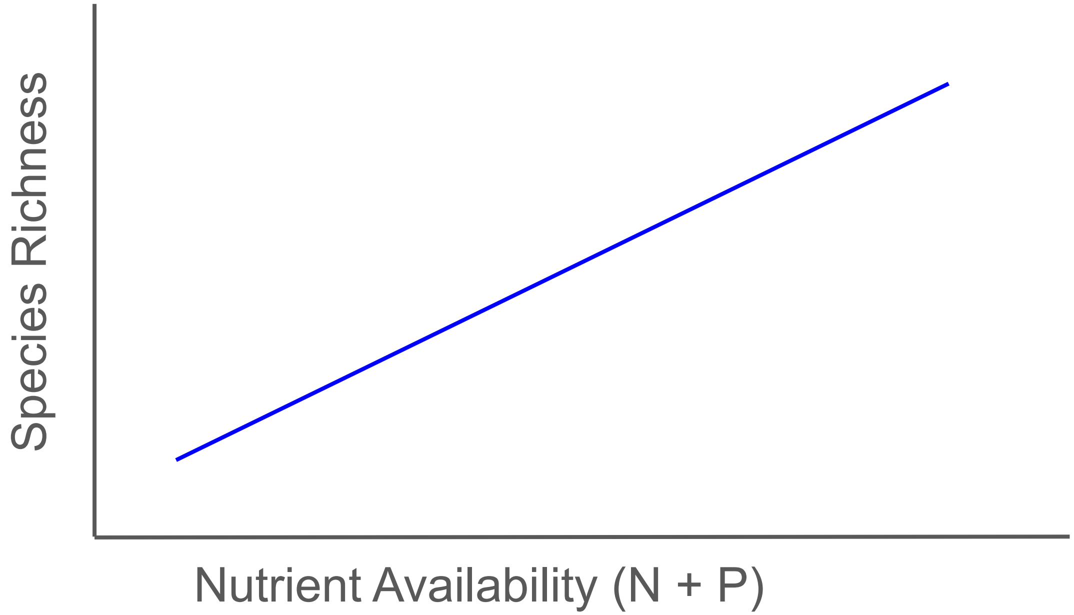
Hypotheses tested

Higher diversity where nutrients aren’t limiting



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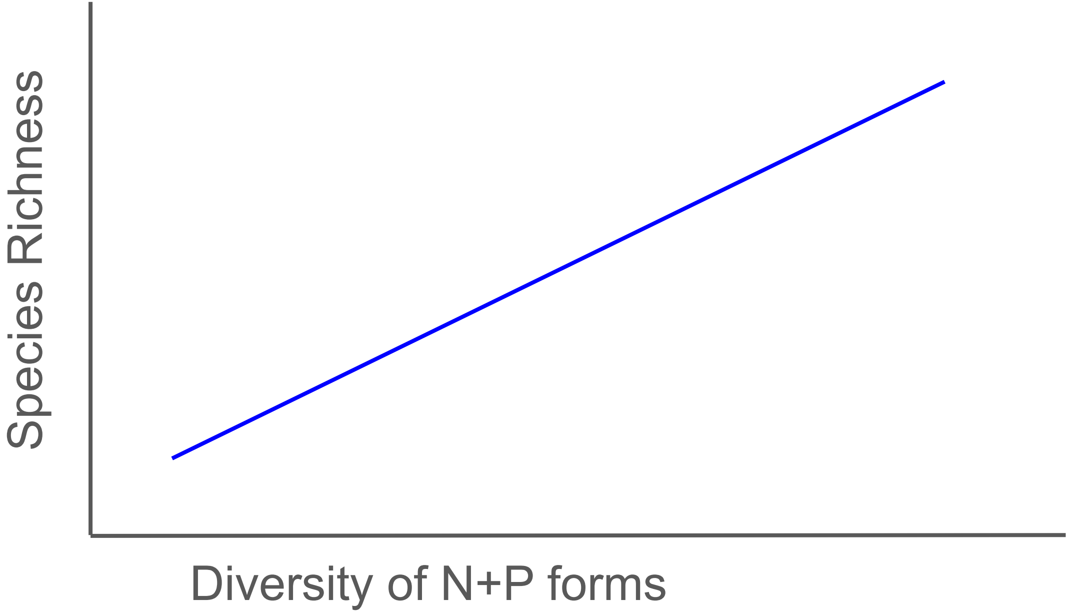
Hypotheses tested

Higher diversity where multiple forms of nutrients exists (partitioning)



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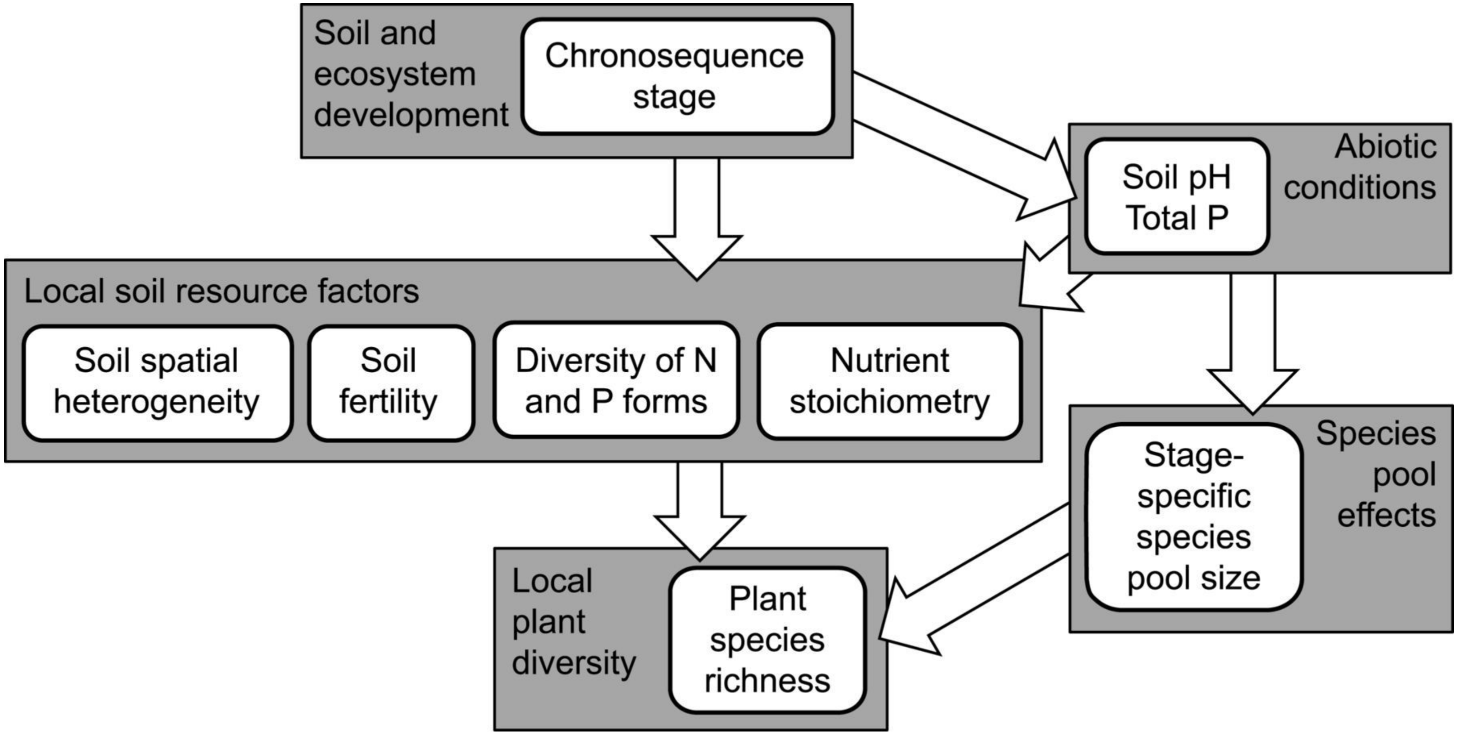

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Methods

- 60 plots (10 m x 10 m)
- 6 age classes (10 plots each)
- In each plot
    - Species
    - Density
    - Cover
    - Soil samples (nutrients and fertility analyses)
    - Leaf sample (for nutrient analyses)
- Species pool estimated:

$$S_{\mathrm{Jack2}} = S_{\mathrm{obs}} + \left[\frac{Q_1(2m-3)}{m} - \frac{Q_2(m-2)^2}{m(m-1)}\right]$$

where $S_{\mathrm{Jack\,2}}$ = extrapolated richness, S_{obs} = observed richness, Q_1 = number of species found in only one sample, Q_2 = number of species found in precisely two samples, and m = number of samples. Nonparametric richness estimation was performed using the ‘vegan’ packages in R (*48*).

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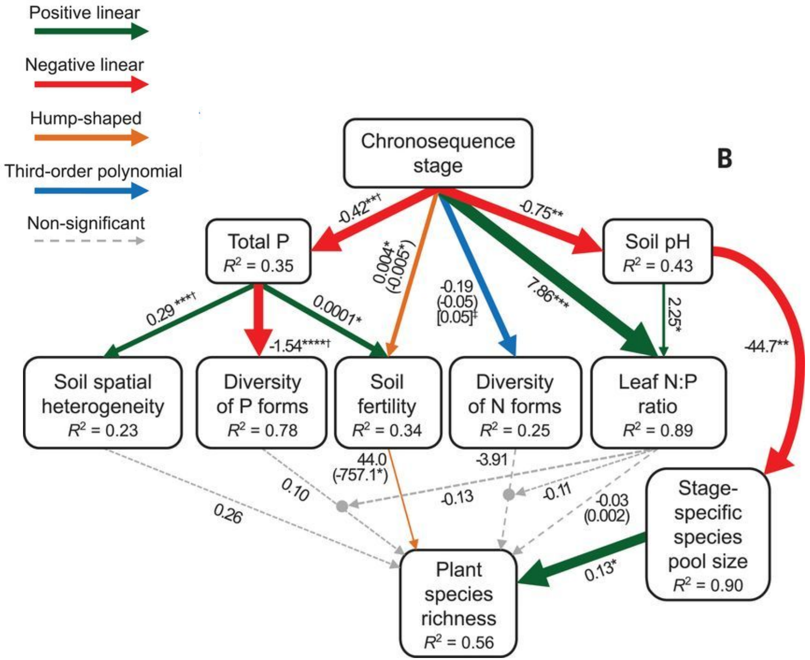
Results



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Plot-level properties didn’t explain plot richness very well

Instead:

- Soil aging changed soil pH
- Different pools of species were specialized on different pH values

Competition and ciliates

ECOLOGY LETTERS

Ecology Letters, (2011) **14**: 782–787 doi: 10.1111/j.1461-0248.2011.01644.x

LETTER

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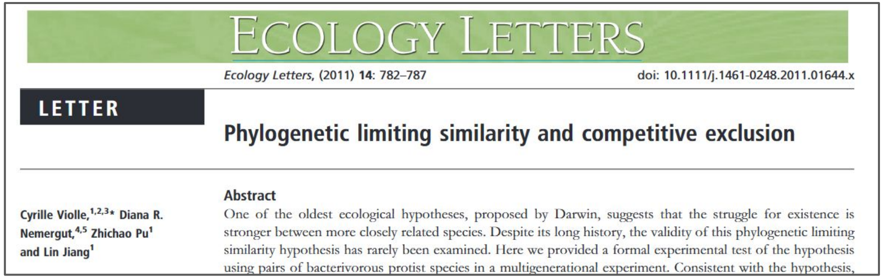

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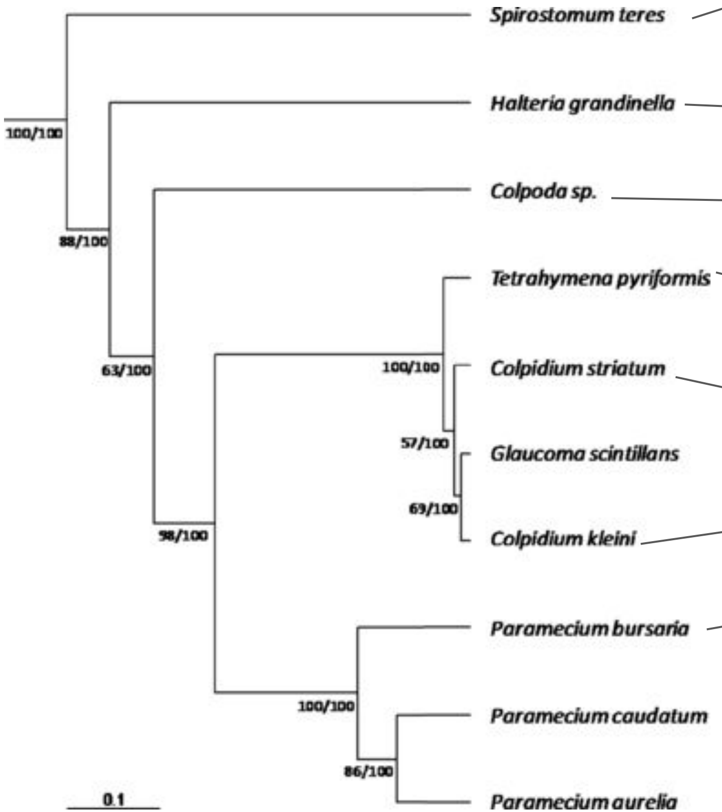

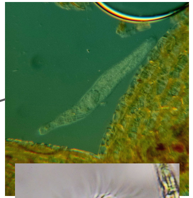

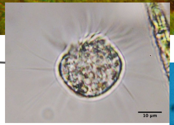

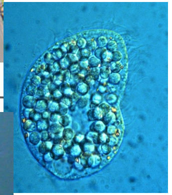

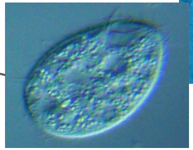

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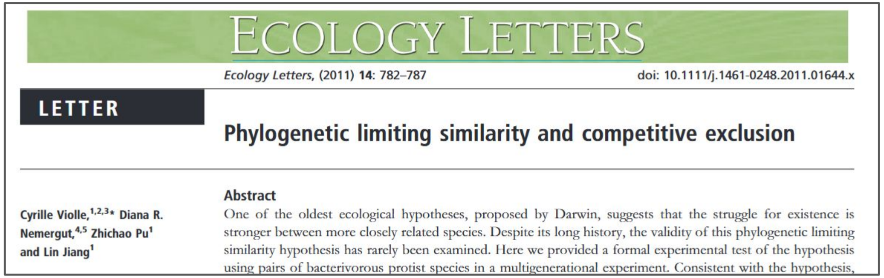
Methods

- Grew populations in pairs and alone
- Grown with 3 different bacteria species (= food)
- Tracked population sizes over time

Competition and ciliates



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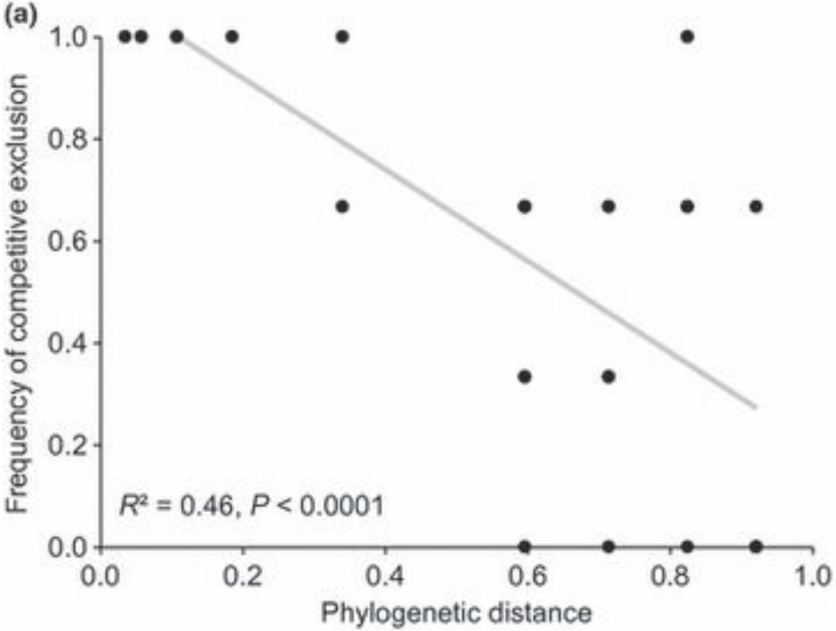
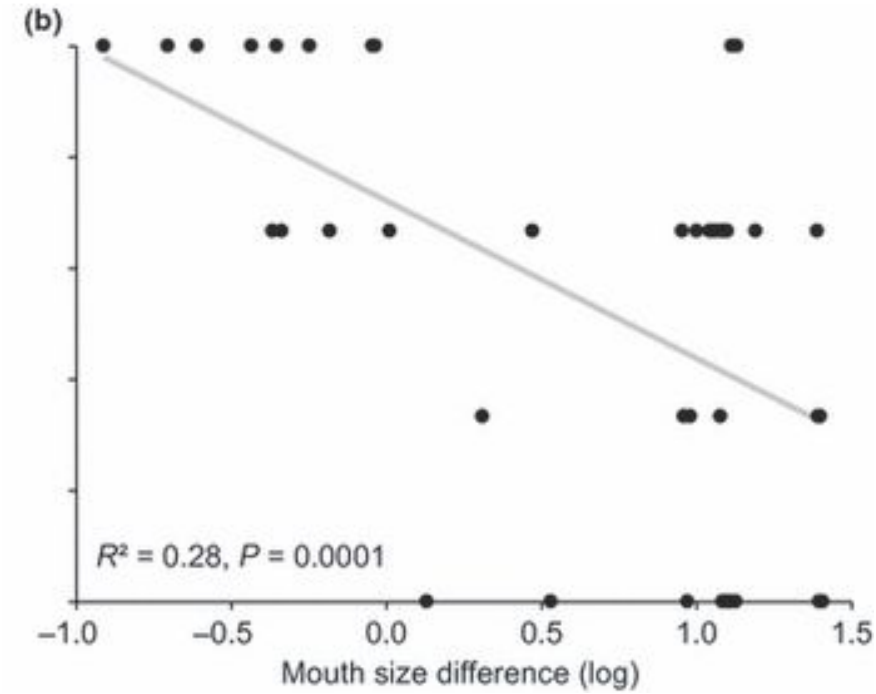
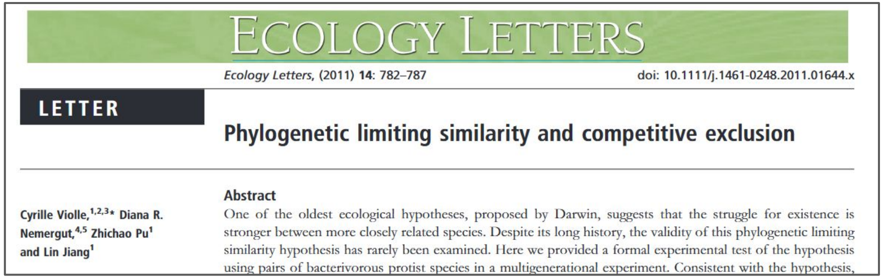


Closely relatives less likely to coexist - suggests competitive exclusion

Competition and ciliates





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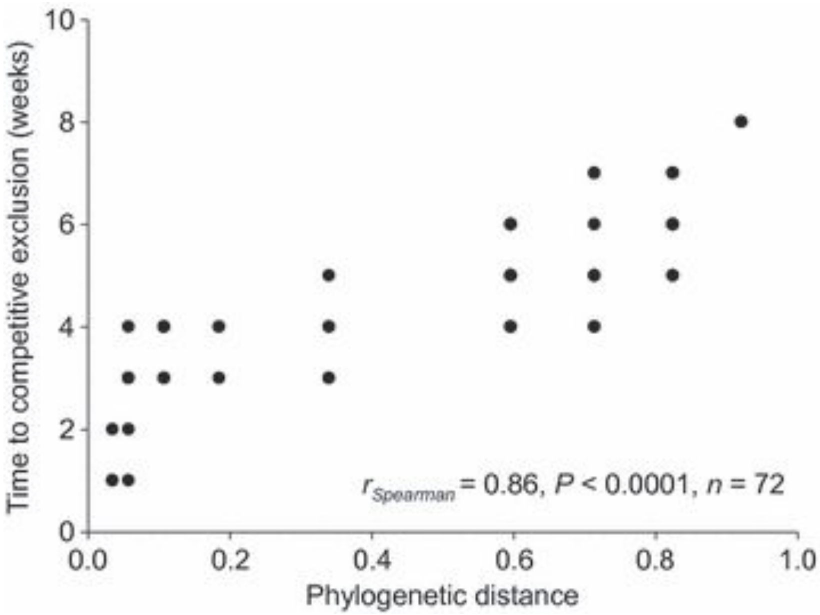
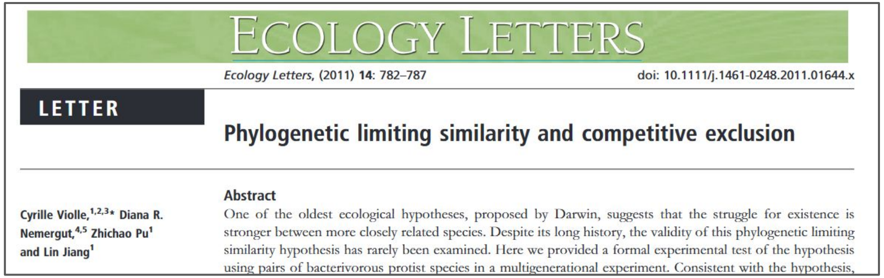
Trait analysis suggests mouth size might play a role

Competition and ciliates

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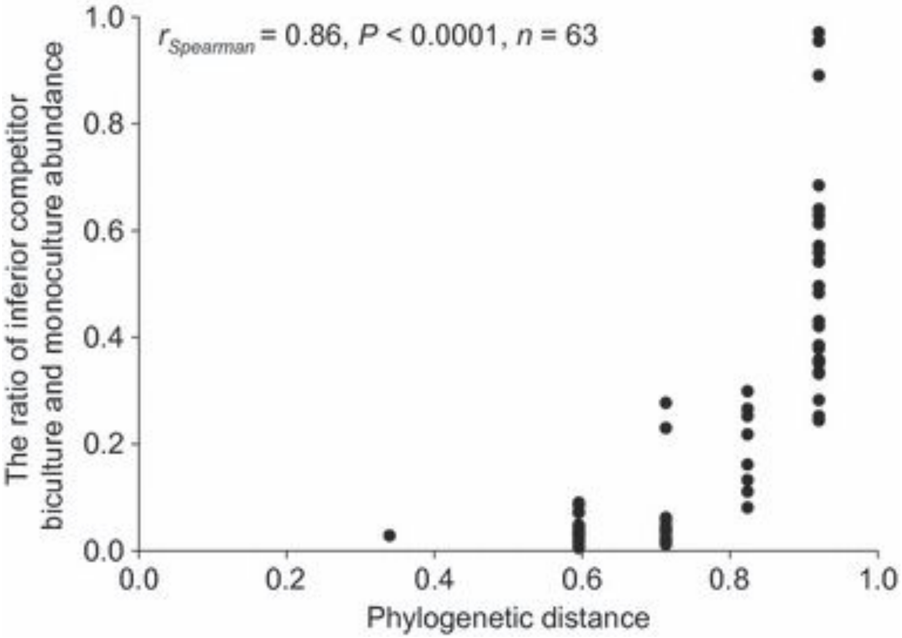
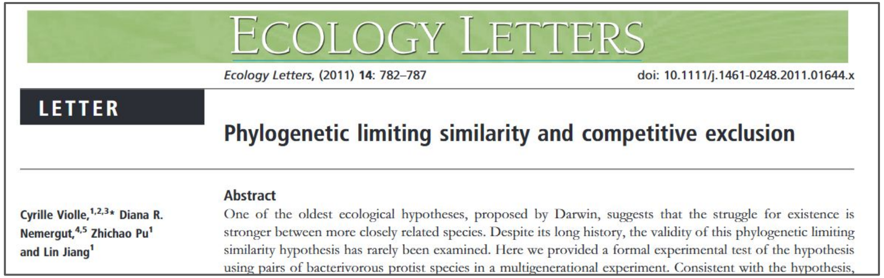
Competitive exclusion happens faster with close relatives

Competition and ciliates





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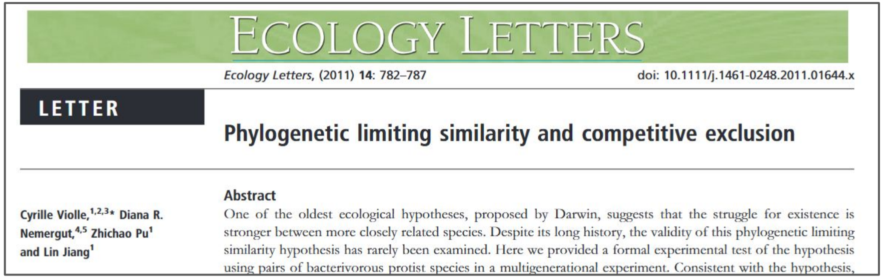
Smaller effects relative to monoculture at larger distances

Competition and ciliates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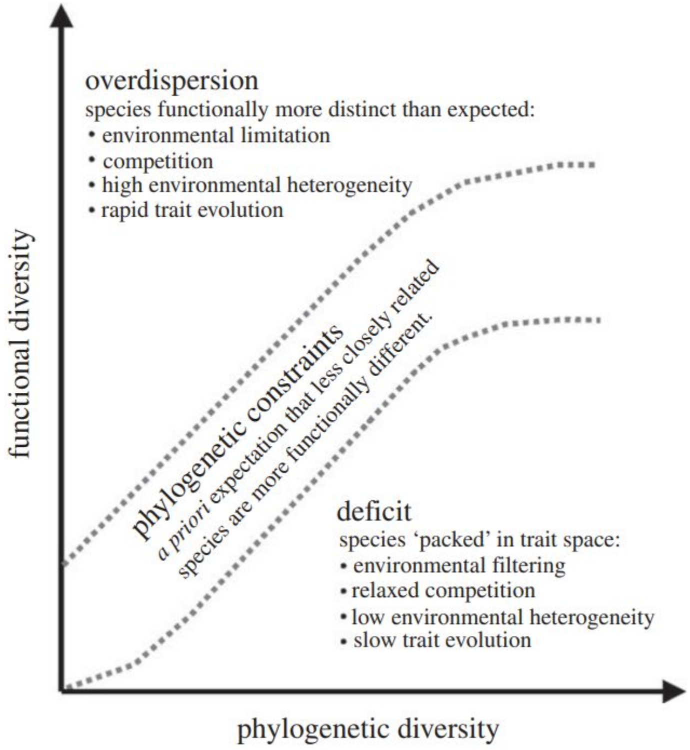
Provides strong, experimental test of competition driving assembly of ciliate communities

- Phylogeny served as a useful predictor
- Seems to be driven by feeding niches

Close relatives

- Faster exclusion
- More likely to exclude each other
- Larger impact on population sizes

Mammal assembly



PHILOSOPHICAL TRANSACTIONS OF THE ROYAL SOCIETY B

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Understanding global patterns of mammalian functional and phylogenetic diversity

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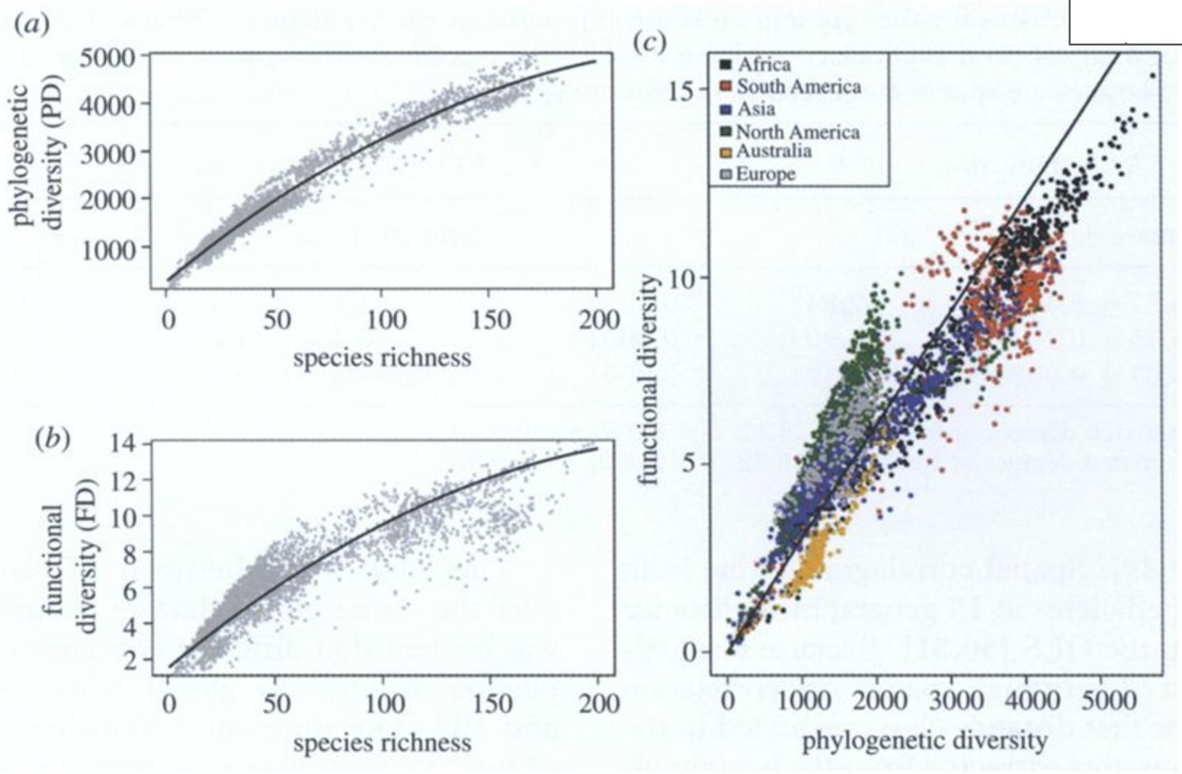

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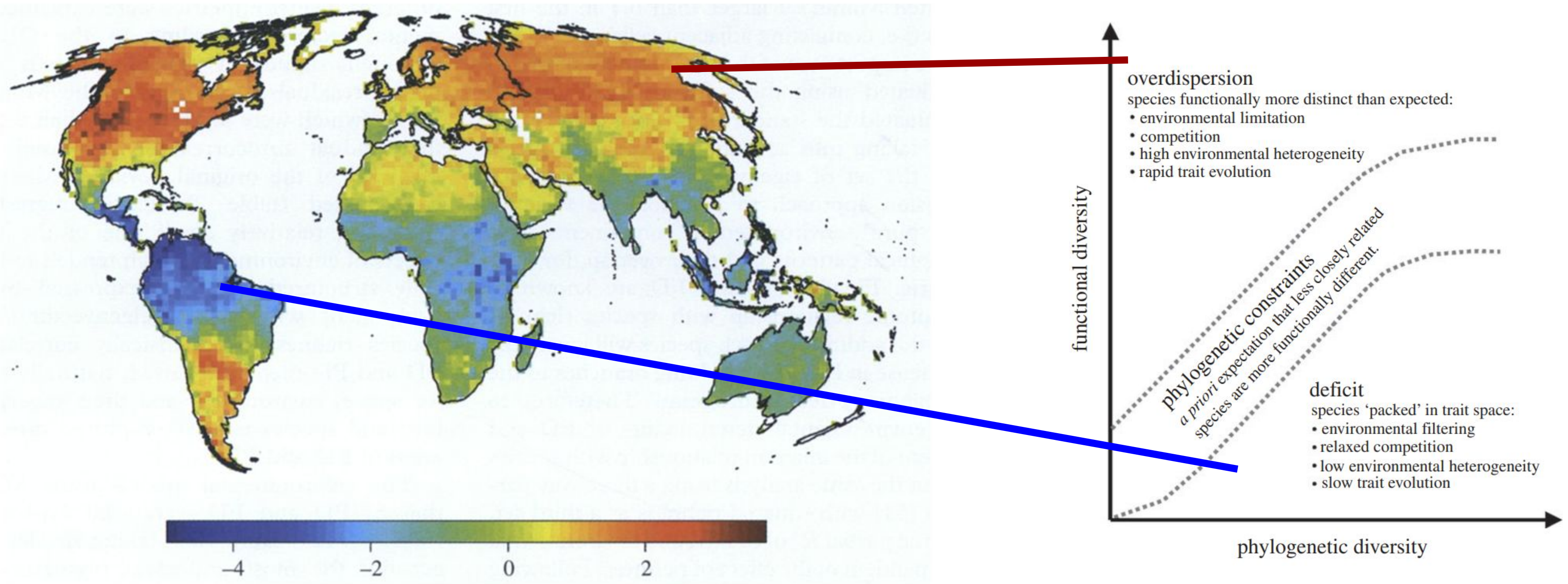

Disease dynamics

ECOLOGY LETTERS

Ecology Letters, (2019) **22**: 138–148 doi: 10.1111/ele.13176

LETTER

Past is prologue: host community assembly and the risk of infectious disease over time

Fletcher W. Halliday,[1,2*] Robert W. Heckman,[1] Peter A. Wilfahrt[3] and Charles E. Mitchell[1,3]

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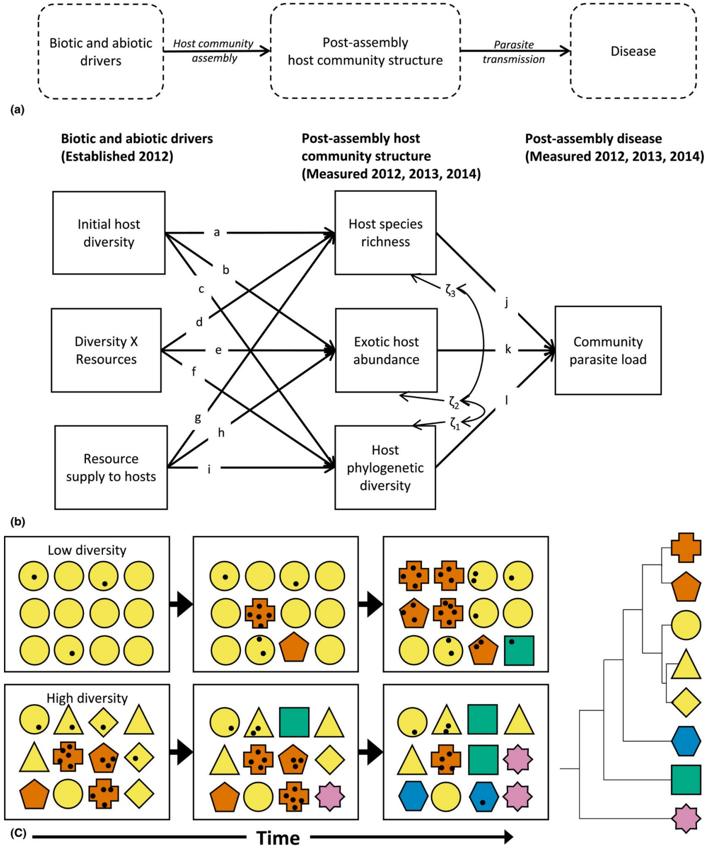
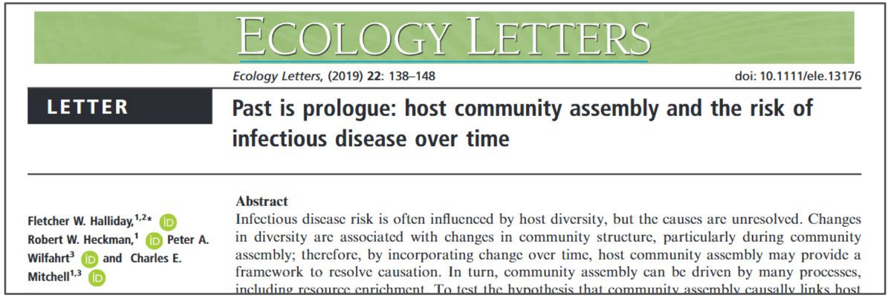
Infectious disease risk is often influenced by host diversity, but the causes are unresolved. Changes in diversity are associated with changes in community structure, particularly during community assembly; therefore, by incorporating change over time, host community assembly may provide a framework to resolve causation. In turn, community assembly can be driven by many processes, including resource enrichment. To test the hypothesis that community assembly causally links host

Disease dynamics





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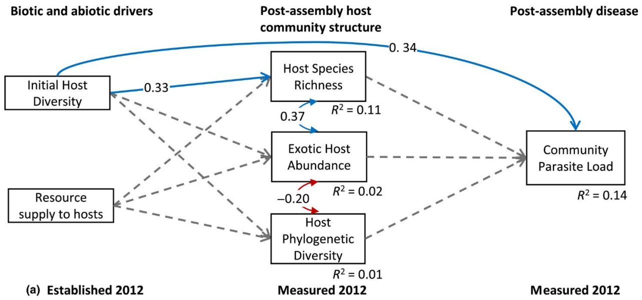
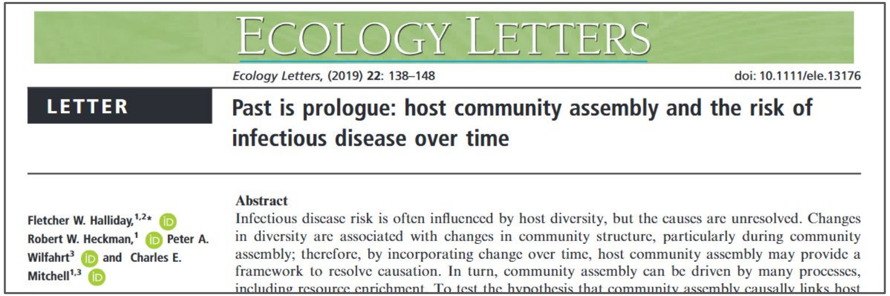
Methods:

- Hosts = plants
- 120 plots (1m x 1 m)
- Treatments:
  - 1 vs 5 species (6 different treatment sets of species each)
  - High vs low resource supply (fertilized vs unfertilized)
- Disease:
  - Subsets of leaves samples per plot per species
  - Parasites identified by symptoms, DNA
  - Calculated parasite load at total leaf area damaged per plot
- Native vs Exotic
  - Followed USDA categorization

Disease dynamics





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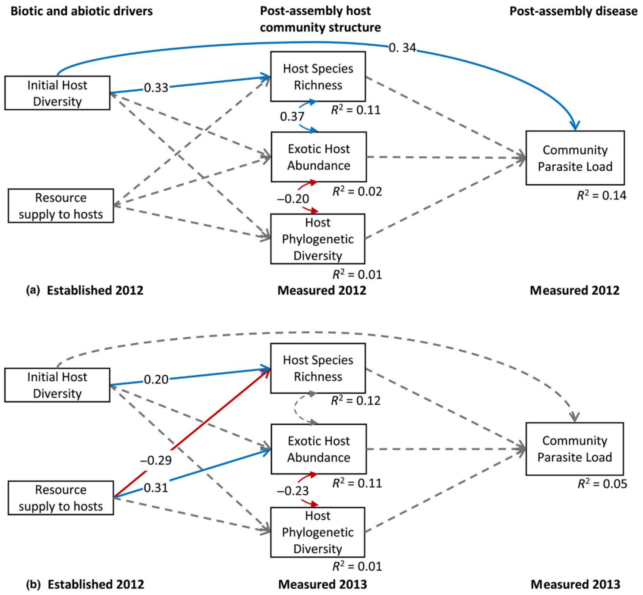

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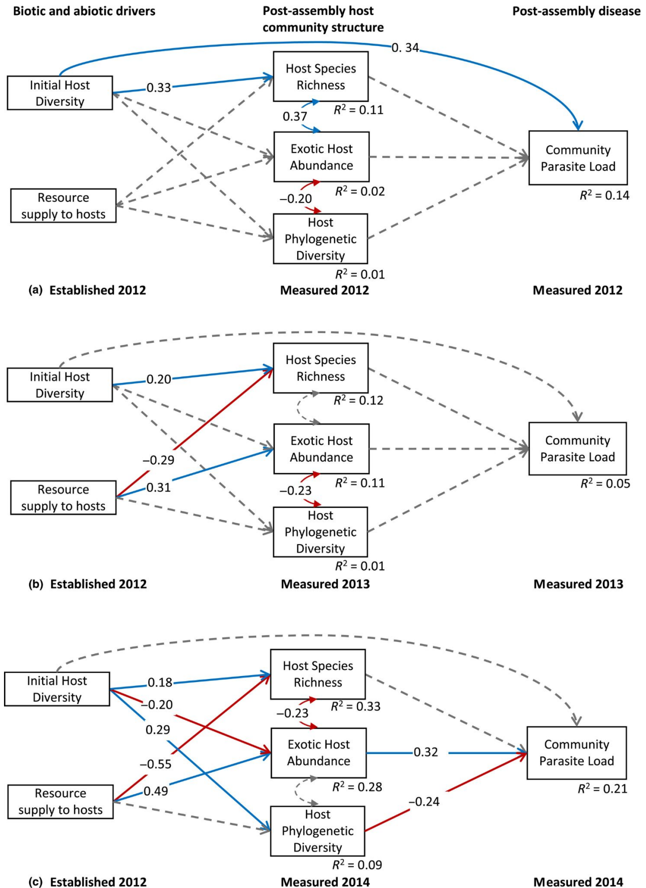

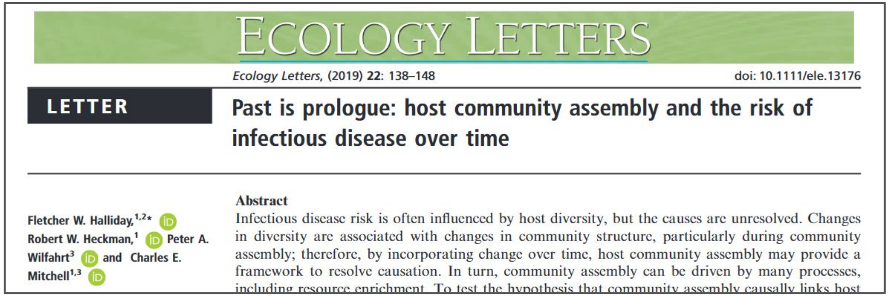

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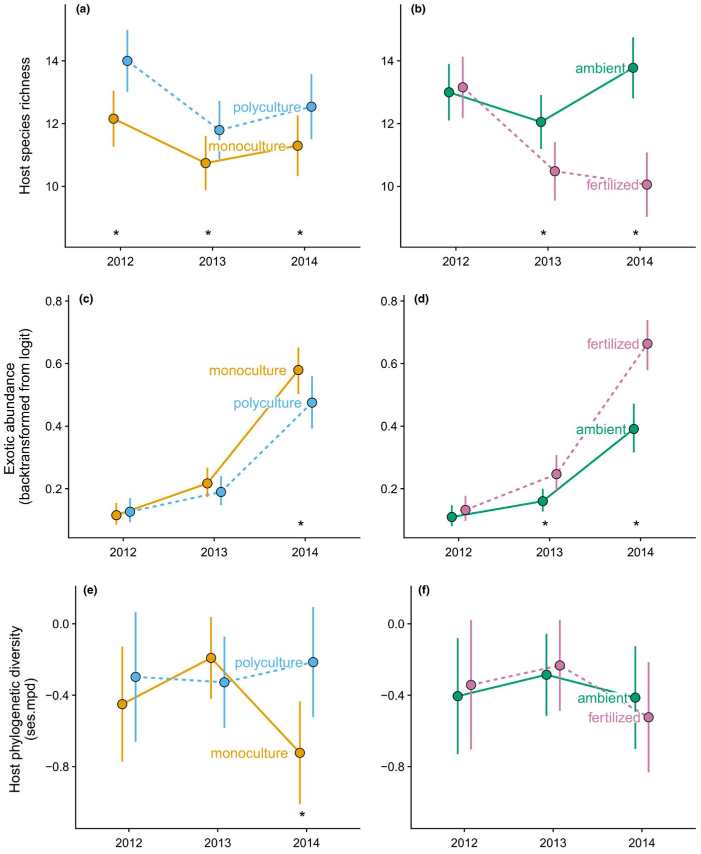

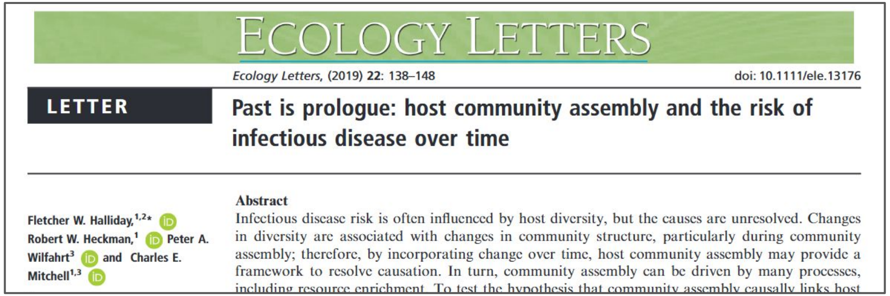

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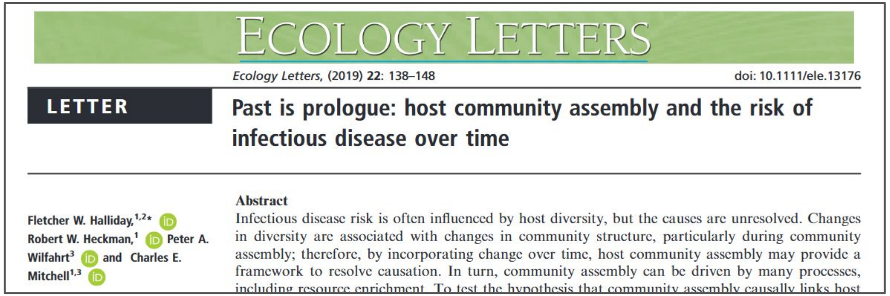
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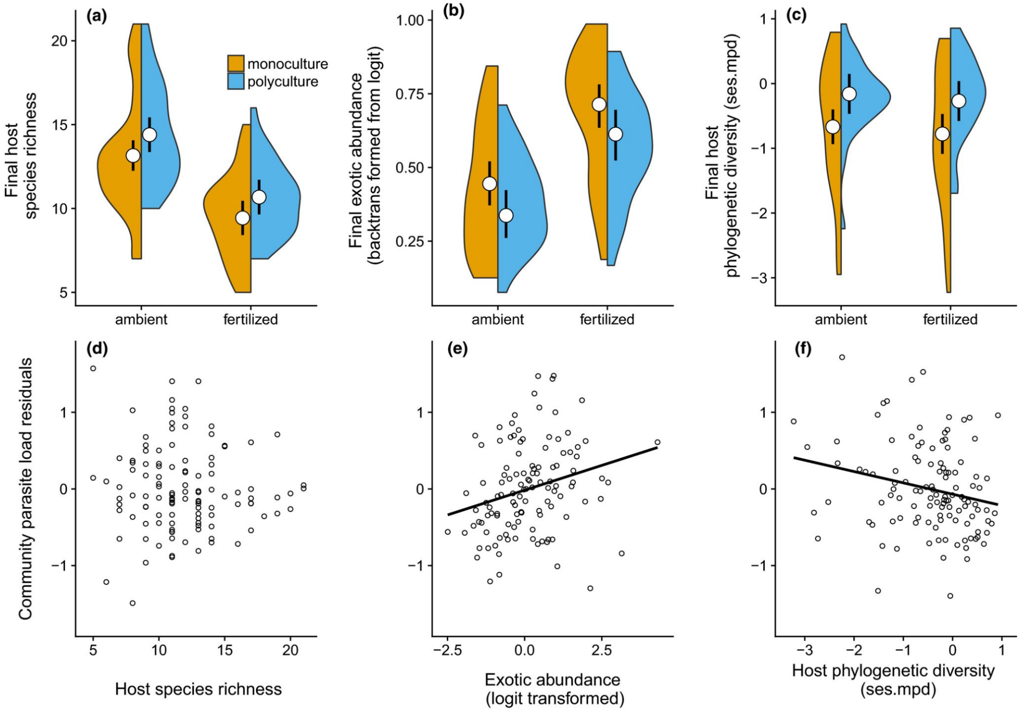

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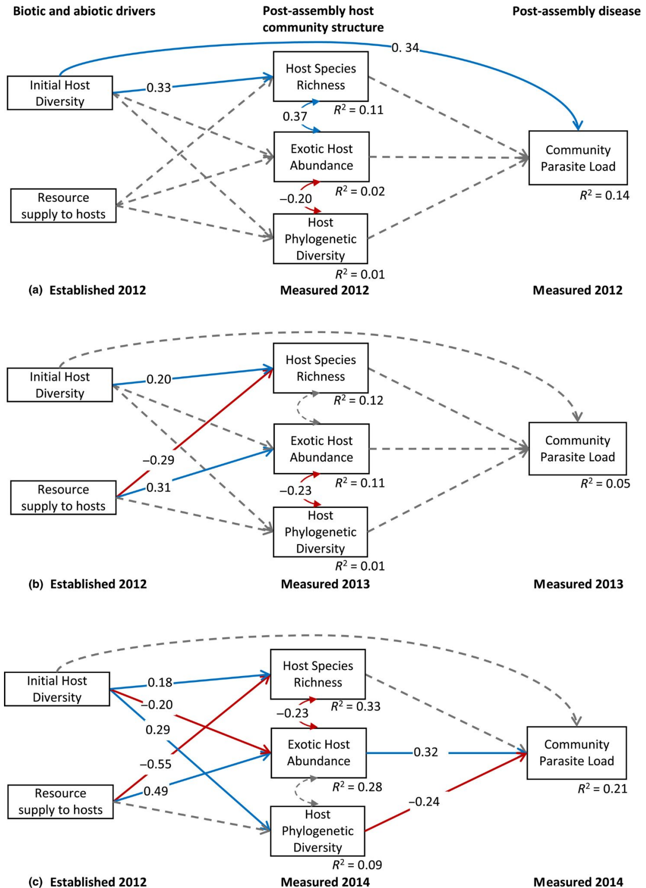

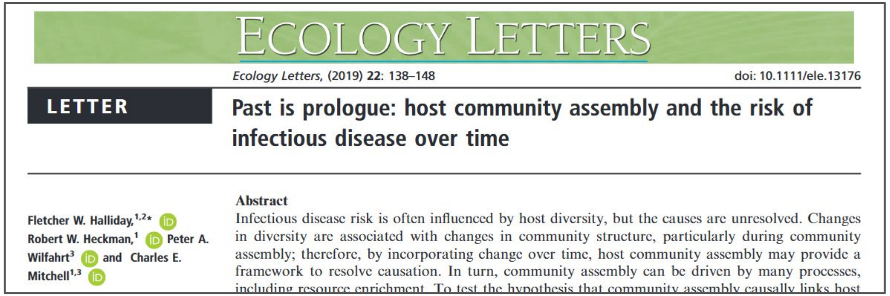

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Community assembly mechanisms can change over time!

- Importance of considering temporal dynamics.
- Just because you don't find evidence of something doesn't mean it won't be important at another time!

Next class: Community assembly and global change